

Targets

Targets are unique operating system identifiers taken from the versions of those specific operating systems which adapt the selected exploit module to run on that particular version of the operating system. The **show targets** command issued within an exploit module view will display all available vulnerable targets for that specific exploit, while issuing the same command in the root menu, outside of any selected exploit module, will let us know that we need to select an exploit module first.

MSF - Show Targets

MSF - Show Targets

```
msf6 > show targets
```

```
[-] No exploit module selected.
```

When looking at our previous exploit module, this would be what we see:

MSF - Show Targets

```
msf6 exploit(windows/smb/ms17_010_psexec) > options
```

Name	Current Setting	Required	Description
----	-----	-----	-----
DBGTRACE	false	yes	Show extra debug trace info
LEAKATTEMPTS	99	yes	How many times to try to leak trans
NAMEDPIPE		no	A named pipe that can be connected
NAMED_PIPES	/usr/share/metasploit-framework/data/wo	yes	List of named pipes to check
RHOSTS	10.10.10.40	yes	The target host(s), see https://github.com/Hackplayers/Exploit-0x00000000/wiki/Using-Metasploit
RPORT	445	yes	The Target port (TCP)
SERVICE_DESCRIPTION		no	Service description to to be used o
SERVICE_DISPLAY_NAME		no	The service display name
SERVICE_NAME		no	The service name
SHARE	ADMIN\$	yes	The share to connect to, can be an
			ormal read/write folder share
SMBDomain	.	no	The Windows domain to use for autho
SMBPass		no	The password for the specified user
SMBUser		no	The username to authenticate as

```
Payload options (windows/meterpreter/reverse_tcp):
```

Name	Current Setting	Required	Description
----	-----	-----	-----
EXITFUNC	thread	yes	Exit technique (Accepted: '', seh, thread, process, none)
LHOST		yes	The listen address (an interface may be specified)
LPORT	4444	yes	The listen port

```
Exploit target:
```

Id	Name
--	----
0	Automatic

Selecting a Target

We can see that there is only one general type of target set for this type of exploit. What if we change the exploit module to something that needs more specific target ranges? The following exploit is aimed at:

- **MS12-063 Microsoft Internet Explorer execCommand Use-After-Free Vulnerability.**

If we want to find out more about this specific module and what the vulnerability behind it does, we can use the **info** command. This command can help us out whenever we are unsure about the origins or functionality of different exploits or auxiliary modules. Keeping in mind that it is always considered best practice to audit our code for any artifact generation or 'additional features', the **info** command should be one of the first steps we take when using a new module. This way, we can familiarize ourselves with the exploit functionality while assuring a safe, clean working environment for both our clients and us.

MSF - Target Selection

MSF - Target Selection

```
msf6 exploit(windows/browser/ie_execcommand_uaf) > info
```

```
Name: MS12-063 Microsoft Internet Explorer execCommand Use-After-Free Vulnerability
Module: exploit/windows/browser/ie_execcommand_uaf
Platform: Windows
Arch:
Privileged: No
License: Metasploit Framework License (BSD)
Rank: Good
Disclosed: 2012-09-14
```

Provided by:

```
unknown
eromang
binjo
sinn3r <sinn3r@metasploit.com>
juan vazquez <juan.vazquez@metasploit.com>
```

Available targets:

```
Id  Name
--  ---
0   Automatic
1   IE 7 on Windows XP SP3
2   IE 8 on Windows XP SP3
3   IE 7 on Windows Vista
4   IE 8 on Windows Vista
5   IE 8 on Windows 7
6   IE 9 on Windows 7
```

Check supported:

```
No
```

Basic options:

Name	Current Setting	Required	Description
OBfuscATE	false	no	Enable JavaScript obfuscation
SRVHOST	0.0.0.0	yes	The local host to listen on. This must be an address on the local machine.
SRVPORT	8080	yes	The local port to listen on.
SSL	false	no	Negotiate SSL for incoming connections
SSLCert		no	Path to a custom SSL certificate (default is randomly generated)
URIPATH		no	The URI to use for this exploit (default is random)

Payload information:

Description:

This module exploits a vulnerability found in Microsoft Internet Explorer (MSIE). When rendering an HTML page, the CMshtmlEd object gets deleted in an unexpected manner, but the same memory is reused again later in the CMshtmlEd::Exec() function, leading to a use-after-free condition. Please note that this vulnerability has been exploited since Sep 14, 2012. Also, note that presently, this module has some target dependencies for the ROP chain to be valid. For WinXP SP3 with IE8, JScript must be present.

chain to be valid. For Windows SP3 with IE8, msvcrt must be present (as it is by default). For Vista or Win7 with IE8, or Win7 with IE9, JRE 1.6.x or below must be installed (which is often the case).

References:

<https://cvedetails.com/cve/CVE-2012-4969/>
OSVDB (85532)
<https://docs.microsoft.com/en-us/security-updates/SecurityBulletins/2012/MS12-063>
<http://technet.microsoft.com/en-us/security/advisory/2757760>
<http://eromang.zataz.com/2012/09/16/zero-day-season-is-really-not-over-yet/>

Looking at the description, we can get a general idea of what this exploit will accomplish for us. Keeping this in mind, we would next want to check which versions are vulnerable to this exploit.

MSF - Target Selection

```
msf6 exploit(windows/browser/ie_execcommand_uaf) > options
```

Module options (exploit/windows/browser/ie_execcommand_uaf):

Name	Current Setting	Required	Description
-----	-----	-----	-----
OBFUSCATE	false	no	Enable JavaScript obfuscation
SRVHOST	0.0.0.0	yes	The local host to listen on. This must be an address on the local machine
SRVPORT	8080	yes	The local port to listen on.
SSL	false	no	Negotiate SSL for incoming connections
SSLCert		no	Path to a custom SSL certificate (default is randomly generated)
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Exploit target:

Id	Name
--	----
0	Automatic

```
msf6 exploit(windows/browser/ie_execcommand_uaf) > show targets
```

Exploit targets:

Id	Name
--	----
0	Automatic
1	IE 7 on Windows XP SP3
2	IE 8 on Windows XP SP3
3	IE 7 on Windows Vista
4	IE 8 on Windows Vista
5	IE 8 on Windows 7
6	IE 9 on Windows 7

We see options for both different versions of Internet Explorer and various Windows versions. Leaving the selection to **Automatic** will let msfconsole know that it needs to perform service detection on the given target before launching a successful attack.

If we, however, know what versions are running on our target, we can use the **set target <index no.>** command to pick a target from the list.

MSF - Target Selection

```
msf6 exploit(windows/browser/ie_execcommand_uaf) > show targets
```

Exploit targets:

Id	Name
--	----
0	Automatic

```
1 IE 7 on Windows XP SP3
2 IE 8 on Windows XP SP3
3 IE 7 on Windows Vista
4 IE 8 on Windows Vista
5 IE 8 on Windows 7
6 IE 9 on Windows 7
```

```
msf6 exploit(windows/browser/ie_execcommand_uaf) > set target 6
```

```
target => 6
```

Target Types

There is a large variety of target types. Every target can vary from another by service pack, OS version, and even language version. It all depends on the return address and other parameters in the target or within the exploit module.

The return address can vary because a particular language pack changes addresses, a different software version is available, or the addresses are shifted due to hooks. It is all determined by the type of return address required to identify the target. This address can be `jmp esp`, a jump to a specific register that identifies the target, or a `pop/pop/ret`. For more on the topic of return addresses, see the [Stack-Based Buffer Overflows on Windows x86](#) module. Comments in the exploit module's code can help us determine what the target is defined by.

To identify a target correctly, we will need to:

- Obtain a copy of the target binaries
- Use msfpescan to locate a suitable return address

Later in the module, we will be delving deeper into exploit development, payload generation, and target identification.

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Metasploit-Framework Updates - August 2020

My Workstation

OFFLINE

 Start Instance

 / 1 spawns left